

Rowing Beyond: A Demonstration of Steering Methods for Rowing-based Locomotion in Virtual Environments

Martin Hedlund

marthed@kth.se

KTH Royal Institute of Technology
Stockholm, Sweden

Gerrit Meixner

gerrit.meixner@hs-heilbronn.de

Heilbronn University
Heilbronn, Germany

Cristian Bogdan

cristi@kth.se

KTH Royal Institute of Technology
Stockholm, Sweden

Andrii Matviienko

andriim@kth.se

KTH Royal Institute of Technology
Stockholm, Sweden



Figure 1: The three steering methods for rowing-based locomotion: (a) head-based steering indicates the direction of movement according to the head's orientation, (b) hands-based steering employs handlebar rotation on a 2D plane and buttons under thumbs to go up and down, and (c) feet-based steering is based on the pressure distribution between left and right feet for steering on a 2D plane and between toes and toes joints for steering in a 3D space.

ABSTRACT

Rowing has great potential in Virtual Reality (VR) exergames as it requires physical effort and uses physical motion to map the locomotion in a virtual space. However, rowing in VR is currently restricted to locomotion along one axis, leaving 2D and 3D locomotion out of the scope. To facilitate rowing-based locomotion, we implemented three steering techniques based on head, hands, and feet movements for 2D and 3D VR environments.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

virtual reality, rowing, exergame, locomotion, steering

ACM Reference Format:

Martin Hedlund, Cristian Bogdan, Gerrit Meixner, and Andrii Matviienko. 2024. Rowing Beyond: A Demonstration of Steering Methods for Rowing-based Locomotion in Virtual Environments. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '24, May 11–16, 2024, Honolulu, HI, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0331-7/24/05

<https://doi.org/10.1145/3613905.3648643>

11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3648643>

1 INTRODUCTION

The use of Virtual Reality (VR) for exergames can increase players motivation to exercise through immersive and engaging experiences [5, 8, 11, 20, 24, 31, 35, 36]. Rowing is a form of physical exercise with great potential in VR exergames since it requires physical effort and uses physical motion to map the locomotion in virtual environments [5, 20, 27]. While previous works on rowing in VR have explored one-dimensional locomotion [3, 25, 28, 29, 32], i.e., forward/backward, rowing-based locomotion in 2D and 3D spaces remains underexplored in virtual environments. Given that virtual environments are infinite and have no restriction of natural laws, on a horizontal 2D plane, users can navigate a virtual sea similar, while in a 3D space, the rowing experience can be transformed into a flying vehicle, or a submarine to steer beyond the surface and gravity. As Ivan Sutherland famously stated: “There is no reason why the objects displayed by a computer have to follow the ordinary rules of physical reality” [30]. However, steering while rowing has no obvious mapping for 2D and 3D spaces, and the rowing activity places physical constraints on bodily movements, e.g., the hands should always hold a handlebar, and feet should be placed at specific locations. Therefore, the question is how rowing-based steering can facilitate locomotion in 2D and 3D space and how efficient they will be. Answering both questions will show the most efficient way to fully explore 2D and 3D virtual environments and facilitate users’

exertion. In this work, we aim to advance rowing-based locomotion on 2D planes and 3D spaces in virtual environments by employing steering methods. For this, we implemented three steering methods based on (1) the physical affordances of a rowing machine, and (2) established steering methods based on head, hands and feet interaction (Figure 1). Head steering facilitates locomotion in the direction of head orientation, hand steering utilizes the rotations of a handlebar, and feet steering employs pressure distribution of toes and big toe joints on feet placeholders. In the following sections we describe our design considerations and implementation details.

2 DESIGN CONSIDERATIONS FOR ROWING-BASED LOCOMOTION

In this section, we introduce design considerations for rowing-based locomotion. These considerations are based on the physical constraints imposed by rowing machines and our aim to facilitate good performance for rowing in an exergame context.

2.1 Metaphor considerations

The locomotion generated by the rowing machine could be mapped to any metaphor in the virtual environment. We opted for a rowing boat metaphor as rowing machines have been substituted for real rowing since their invention [13]. We generally aimed for a realistic “feel” in which users’ naive sense of physics helps them understand the interactions [16], e.g., pulling harder increases steering rotation force. However, we also went beyond realism for better ergonomics and efficient locomotion [1]. The virtual locomotion speed was increased compared to a realistic rowing boat since we believe this facilitated a more engaging experience, an important motivational aspect of frequent exergame usage [22]. We also assumed moving forward instead of backward (real rowing is performed backward) would be more engaging and ergonomic in most games. Lastly, 3D “flying” locomotion for boats is unrealistic. Still, we believed it would be easier for users to have the same metaphor and controls in both 2D and 3D instead of introducing a new metaphor, e.g., airplane, helicopter.

2.2 Locomotion considerations

2.2.1 When to provide steering power. According to Bowman, when and how input is triggered is one of three important characteristics of any locomotion technique, together with velocity and direction selection [6]. Although steering power could technically be available as soon as the user provides input, this would feel unrealistic for the user if a rowing motion is not co-occurring. Rowing differs from other vehicle steering methods because it requires the oars in the water. In other words, even if the virtual boat moves forward and the oars are up, steering is impossible. While there are no oars on a rowing machine, we decided to provide steering power when the users were pulling the handlebar towards themselves as it mimics the feeling of putting the oars in the water and pulling.

2.2.2 How to control steering power. Various possibilities exist for controlling the applied amount of turning force [9]. Other locomotion methods have increased the force based on the input frequency per time unit, e.g., increasing input events during a second increases the amount of force. This works if a user repeats an input action

frequently, such as “hammering” a button, which allows a user to easily prevent further input by not acting [6]. In contrast, continuous input directly sets the turn force value from the input source. We decided to provide continuous input so the user could more smoothly set the value, thereby avoiding distracting maneuvers. This means that unintended inputs could be a problem, making the steering feel “wobbly”. However, we avoided this by setting continuous input values only during pulling. We amplified the steering input with a power function [9] with pulling speed as an input, causing faster, heavier pulls to increase the turning force exponentially more than slower pulls.

2.2.3 How to move in 3D space. We considered two forms of forces for movement through vertical space: *combined* and *separate*. A combined force would only apply speed power on the forward vector of the virtual rowing boat. This implies that the angle of the virtual boat must be pitched for locomotion to occur. This creates a mismatch between the user’s physical and visual orientations in VR, which increases the risk of VR sickness [14]. Roll and yaw maneuvers must also be implemented for a combined force, increasing steering complexity. Instead, we decided on a separate force approach in which up- and downward input creates vertical locomotion directly perpendicular to the forward vector. To avoid an elevator-like experience, we multiplied the vertical force with the handlebar pull force. Thus, to move vertically, forward speed is needed. We did not include gravity to allow users to rest at higher altitudes. To provide a fluent experience we made sure that 2D and 3D steering actions could be performed simultaneously, e.g. moving down and left at the same time.

2.3 Body-specific considerations

We implemented three steering methods based on head, hands, and feet input. Steering techniques are commonly categorized based on which body part controls input [7, 17, 21, 26]. The most frequently used techniques uses input from *head*, *torso* and *hands* [12, 17], with *legs* [4, 37], and *feet* [23, 33, 34] gaining research interest in recent year. A rowing machine restricts users from rotating their torso and repositioning their legs, so we did not implement steering alternatives based on these inputs. The three steering methods we implemented thereby represent the remaining body-specific techniques. Figure 2 illustrates the methods.

2.3.1 Head. For head-based steering, we employed pitch and yaw orientation. We did not consider the headset position as an input because the rowers’ motion on the machine would make it unfeasible. Thus, we mapped the left-right steering to the user’s yaw orientation, i.e., rotating around the upward axis, and vertical steering to the user’s pitch orientation, i.e., looking up and down. An advantage of this method is that it does not require other devices than the VR headset, thus lowering the usage barrier [26].

2.3.2 Hands. Hand-based steering is restricted by the user’s occupation of the hands holding the handlebar. Instead, the handlebar can provide input through 3DoF orientation as with head-based steering [26]. In practice, however, only the handlebar roll and yaw orientations are feasible to use, as the chain attached to the handlebar restricts rotation along the pitch axis. Both roll and yaw are intuitive candidates for left-right steering, resembling the steering

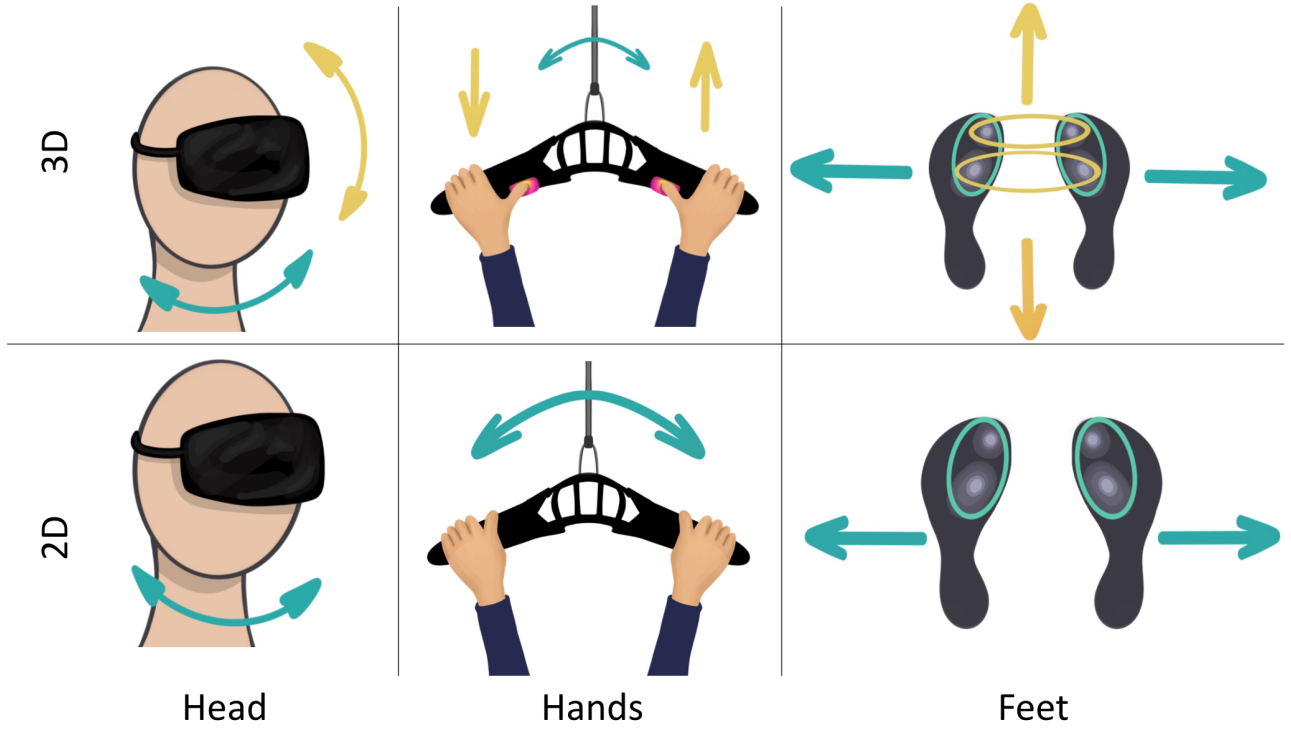


Figure 2: Overview of three steering methods for 2D (lower row) and 3D (upper row) using head, hands, and feet. The head steering facilitates locomotion in the direction of head orientation, hand steering utilizes the rotations of a handlebar, and feet steering employs pressure distribution of toes and big toe joints on feet placeholders.

of a car or bike. While using yaw would cause the users to pull with either the left or right arm mainly, the roll axis affords both arms to be equally involved. This is important for a regular rowing motion, which prevents injuries [32]. We, therefore, used the roll axis of the handlebar for 2D steering. Since the pitch axis is unavailable for 3D steering, other input sources than handlebar orientation had to be used. As the user’s fingers are available, buttons can be attached to the handlebar. We decided to add one button for each thumb, right for up and left for down. The logic behind this mapping is based on possible user familiarity with scooters and mopeds, in which acceleration is typically on the right side. This meant fixed discrete input each frame a button was pushed down. The hands method is arguably most akin to steering a vehicle, e.g., bicycle [18, 19], car [26], or scooter, and could therefore feel familiar to users.

2.3.3 Feet. The user’s feet need to be strapped on the rowing machine footstanch (figure 1) to provide resistance force while pulling, and could therefore not be lifted. This prevents direction from being set by foot orientation or relative foot position [34]. However, the resistance force could be achieved if the heels remained on the footstanch. By loosening the straps a little, the user can elevate one foot a bit, thereby easily controlling the amount of pressure on each foot stance. We used this difference in foot pressure between footstanchs as input for the left or right directions. For 3D steering, the same method can be applied, but the difference must be

measured between the front and back of the feet. Since heels must always apply pressure on the foot stance, we used the difference between the big toes and the big toe joints of the feet for up and down input. Pressing with the joints increases the upward force while pressing with the big toes increases the downward force. An inverse mapping could also be feasible. The logic for this mapping was that pressing on one’s toes makes one fall forward, i.e., pitching down, and pressing on the joints of one’s feet would make one fall backward, i.e., pitching up. By using feet-based steering, hands can be free to perform other actions (e.g. using thumbs for input in some game). The user’s head is also decoupled from the travel direction, allowing the user to more easily view the environment while moving.

3 IMPLEMENTATION

Based on our design considerations, we developed a VR rowing environment¹ where users can navigate a virtual ocean and compete on coin collection tracks. The implementation includes a physical rowing machine, external hardware sensors, and a Virtual Reality environment.

¹https://github.com/marthed/vr_rowing

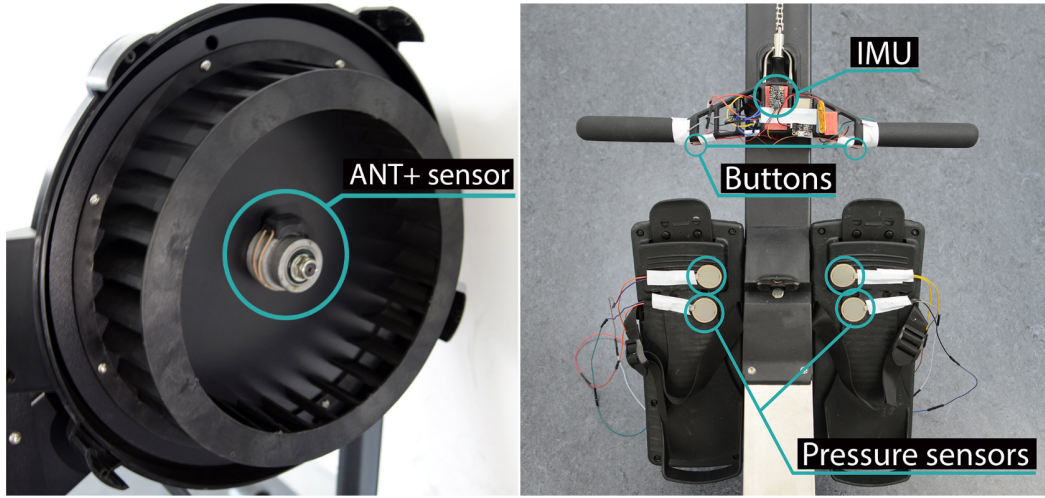


Figure 3: (Left) ANT+ sensor attached to the flywheel for speed input. (Right) Steering sensors for hands (inertial measurement unit, buttons) and feet method (pressure sensors).

3.1 VR Environment

The VR environment was implemented using Unity SDK (2021.1.3) with OculusVR assets and consisted of an ocean with islands. To avoid inconveniences caused by wires, we used the wireless headset Oculus Quest 2. As we later evaluated our steering methods, we designed the virtual environment with minimal distractions, featuring only a flat blue ocean to minimize the impact of VR sickness [2]. Mountains in the distance were added to aid height perception. We also added coin collection tracks for the later user study to compare performance between the steering methods (see study details in [10]).

3.2 Velocity

Force was applied to make the virtual boat move forward by pulling the handlebar of the rowing machine. We used an **Epsilon RX90**² as the physical rowing machine. We fitted the flywheel of the rowing machine with a Garmin Speed Sensor 2, which transmits real-time speed via ANT+ and Bluetooth to the simulation (Figure 3). The continuous value from the speed sensor was applied to the virtual boat's forward direction as a force.

3.3 Direction

To control 2D direction, steering input was applied as a torque rotation force on the virtual boat's yaw-axis. For 3D, we applied an upward/downward force instead of axis rotation to avoid problems described in 2.2.3.

3.3.1 Head-based. The *head-based* method was implemented using the continuous orientation of the Oculus Quest 2 headset (angle tracking error $<0.001^\circ$ [15]) to capture the head's yaw rotation in 2D (sideways), and pitch rotation in 3D (up/down).

3.3.2 Hands-based. For the *hands-based* method, we placed a **LSM6DSOX 6DOF Inertial measurement unit (IMU)** on the handlebar of the rowing machine to continuously measure the roll-axis rotation angle of the handlebar left and right 3. The IMU was connected to a **ESP32-feather** board, which forwarded the IMU signal to the Oculus Headset over a WiFi connection using UDP. A 3.7v 1200mAh ion polymer battery powered the board. Given that turning a handlebar upwards and downwards is difficult and unnatural, as discussed above in the considerations section, we added two buttons (discrete input) on both sides of the handlebar. We aimed to facilitate simultaneous upward-downward and side-wise movement. Pressing the button on the right enables upward movement and the one on the left downwards.

3.3.3 Feet-based. Finally, four **pressure sensors (continuous input, Alpha MF01A-N-221-A01, 0.3-10NF)** on the two footstanchions enabled the *feet-based* method (see 3. Two sensors were placed under the big toes (toe sensors) and two – under the joints connecting the big toes to the foot (joint sensors). Simultaneous pressure on toe sensors facilitates going downwards, and simultaneous pressure on toe joints – upwards. A pressure on a right toe and/or joint sensor enables turning right, and a pressure on a left toe and/or joint sensor – turning left. We used the same type of board, battery, and communication with the headset for the feet sensors.

4 CONCLUSION

In this paper, we demonstrate our implementation of rowing-based locomotion in virtual environments. For this we introduce three steering methods are based on established interaction techniques using head-, hands- and feet-based interaction. Unlike existing rowing-based locomotion approaches, our implementation goes beyond one-dimensional forward motion, and allow travel in both 2D and 3D, opening new exergame design possibilities.

²https://www.sportig.no/pub_docs/files/dokument_sportig/Epsilon-RX90-Manual-EN.pdf

REFERENCES

- [1] Parastoo Abtahi, Sidney Q. Hough, James A. Landay, and Sean Follmer. 2022. Beyond Being Real: A Sensorimotor Control Perspective on Interactions in Virtual Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3491102.3517706>
- [2] Samuel Ang and John Quarles. 2022. You're in for a Bumpy Ride! Uneven Terrain Increases Cybersickness While Navigating with Head Mounted Displays. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 428–435. <https://doi.org/10.1109/VR51125.2022.00062> ISSN: 2642-5254.
- [3] Sebastian Arndt, Andrew Perki, and Jan-Niklas Voigt-Antons. 2018. Using Virtual Reality and Head-Mounted Displays to Increase Performance in Rowing Workouts. In *Proceedings of the 1st International Workshop on Multimedia Content Analysis in Sports (MMSports'18)*. Association for Computing Machinery, New York, NY, USA, 45–50. <https://doi.org/10.1145/3265845.3265848>
- [4] Kenan Bektaş, Tyler Thrash, Mark A. van Raai, Patrik Künzler, and Richard Hahnloser. 2021. The systematic evaluation of an embodied control interface for virtual reality. *PLOS ONE* 16, 12 (Dec. 2021), e0259977. <https://doi.org/10.1371/journal.pone.0259977> Publisher: Public Library of Science.
- [5] Felix Born, Adrian Rygula, and Maic Masuch. 2021. Motivating Players to Perform an Optional Strenuous Activity in a Virtual Reality Exergame Using Virtual Performance Augmentation. *Proceedings of the ACM on Human-Computer Interaction* 5, CHI PLAY (2021), 225:1–225:21. <https://doi.org/10.1145/3474652>
- [6] D.A. Bowman, D. Koller, and L.F. Hodges. 1997. Travel in immersive virtual environments: an evaluation of viewpoint motion control techniques. In *Proceedings of IEEE 1997 Annual International Symposium on Virtual Reality*. 45–52. <https://doi.org/10.1109/VRRAIS.1997.583043> ISSN: 1087-8270.
- [7] Heni Cherni, Natacha Métayer, and Nicolas Souliman. 2020. Literature review of locomotion techniques in virtual reality. *International Journal of Virtual Reality* 20, 1 (March 2020), 1–20. <https://doi.org/10.20870/IJVR.2020.20.1.3183> Number: 1.
- [8] Sebastian Cmentowski, Sukran Karaosmanoglu, Lennart E. Nacke, Frank Steinicke, and Jens Harald Krüger. 2023. Never Skip Leg Day Again: Training the Lower Body with Vertical Jumps in a Virtual Reality Exergame. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–18. <https://doi.org/10.1145/3544548.3580973>
- [9] BoYu Gao, Zijun Mai, Huawei Tu, and Henry Been-Lirn Duh. 2023. Effects of Transfer Functions and Body Parts on Body-Centric Locomotion in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* 29, 8 (Aug. 2023), 3670–3684. <https://doi.org/10.1109/TVCG.2022.3169222> Conference Name: IEEE Transactions on Visualization and Computer Graphics.
- [10] Martin Hedlund, Cristian Bogdan, Gerrit Meixner, and Andrii Matviienko. 2024. Rowing Beyond: Investigating Steering Methods for Rowing-based Locomotion in Virtual Environments. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3613904.3642192>
- [11] Martin Hedlund, Adam Jonsson, Cristian Bogdan, Gerrit Meixner, Elin Ekblom Bak, and Andrii Matviienko. 2023. BlockyVR: Exploring Block-based Programming in Virtual Reality. In *Proceedings of the 22nd International Conference on Mobile and Ubiquitous Multimedia (MUM '23)*. Association for Computing Machinery, New York, NY, USA, 257–269. <https://doi.org/10.1145/3626705.3627779>
- [12] Martin Hedlund, Anders Lundström, Cristian Bogdan, and Andrii Matviienko. 2023. Jogging-in-Place: Exploring Body-Steering Methods for Jogging in Virtual Environments. In *Proceedings of the 22nd International Conference on Mobile and Ubiquitous Multimedia (MUM '23)*. Association for Computing Machinery, New York, NY, USA, 377–385. <https://doi.org/10.1145/3626705.3627778>
- [13] Conor Heffernan. 2016. The History of the Indoor Rower - Physical Culture Study. <https://physicalculturestudy.com/2016/05/06/the-history-of-the-indoor-rower/>, <https://physicalculturestudy.com/2016/05/06/the-history-of-the-indoor-rower/>
- [14] Lawrence J. Hettinger, Kevin S. Berbaum, Robert S. Kennedy, William P. Dunlap, and Margaret D. Nolan. 1990. Vection and Simulator Sickness. *Military Psychology* 2, 3 (Sept. 1990), 171–181. https://doi.org/10.1207/s15327876mp0203_4 Publisher: Routledge. eprint: https://doi.org/10.1207/s15327876mp0203_4
- [15] Valentin Holzwarth, Joy Gisler, Christian Hirt, and Andreas Kunz. 2021. Comparing the Accuracy and Precision of SteamVR Tracking 2.0 and Oculus Quest 2 in a Room Scale Setup. In *2021 the 5th International Conference on Virtual and Augmented Reality Simulations (ICVARs 2021)*. Association for Computing Machinery, New York, NY, USA, 42–46. <https://doi.org/10.1145/3463914.3463921>
- [16] Robert J.K. Jacob, Audrey Girouard, Leanne M. Hirshfield, Michael S. Horn, Orit Shaer, Erin Treacy Solovey, and Jamie Zigelbaum. 2008. Reality-based interaction: a framework for post-WIMP interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08)*. Association for Computing Machinery, New York, NY, USA, 201–210. <https://doi.org/10.1145/1357054.1357089>
- [17] Esteban Segarra Martinez, Annie S. Wu, and Ryan P. McMahan. 2022. Research Trends in Virtual Reality Locomotion Techniques. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 270–280. <https://doi.org/10.1109/VR51125.2022.00046> ISSN: 2642-5254.
- [18] Andrii Matviienko, Florian Müller, Marcel Zickler, Lisa Alina Gasche, Julia Abels, Till Steinert, and Max Mühlhäuser. 2022. Reducing Virtual Reality Sickness for Cyclists in VR Bicycle Simulators. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3491102.3501959>
- [19] Jeremy McDade, Allison Jing, Tasha Stanton, and Ross Smith. 2023. Assessing Superhuman Speed as a Gamified Reward in a Virtual Reality Bike Exergame. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3544549.3585730>
- [20] Daniel J. McDonough, Zachary C. Pope, Nan Zeng, Wenxi Liu, and Zan Gao. 2020. Comparison of College Students' Blood Pressure, Perceived Exertion, and Psychosocial Outcomes During Virtual Reality, Exergaming, and Traditional Exercise: An Exploratory Study. *Games for Health Journal* 9, 4 (Aug. 2020), 290–296. <https://doi.org/10.1089/g4h.2019.0196> Publisher: Mary Ann Liebert, Inc., publishers.
- [21] Ryan P McMahan, Regis Kopper, and Doug A. Bowman. 2014. Principles for Designing Effective 3D Interaction Techniques. In *Handbook of Virtual Environments* (2 ed.), Kelly S. Hale Stanney, Kay M. (Ed.). CRC Press. Num Pages: 28.
- [22] Florian 'Floyd' Mueller, Darren Edge, Frank Vetere, Martin R. Gibbs, Stefan Agamanolis, Bert Bongers, and Jennifer G. Sheridan. 2011. Designing sports: a framework for exertion games. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, Vancouver BC Canada, 2651–2660. <https://doi.org/10.1145/1978942.1979330>
- [23] Florian Müller, Daniel Schmitt, Andrii Matviienko, Dominik Schön, Sebastian Günther, Thomas Kosch, and Martin Schmitz. 2023. TieTacToes: Assessing Toe Movements as an Input Modality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3544548.3580954>
- [24] Taiwoo Park, Uichin Lee, Scott MacKenzie, Miri Moon, Inseok Hwang, and June-hwa Song. 2014. Human factors of speed-based exergame controllers. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1865–1874. <https://doi.org/10.1145/2556288.2557091>
- [25] Brett J. Parton and David L. Neumann. 2019. The effects of competitiveness and challenge level on virtual reality rowing performance. *Psychology of Sport and Exercise* 41 (March 2019), 191–199. <https://doi.org/10.1016/j.psychsport.2018.06.010>
- [26] Bernhard E. Riecke, Joseph J. LaViola, and Ernst Kruijff. 2018. 3D user interfaces for virtual reality and games: 3D selection, manipulation, and spatial navigation. In *ACM SIGGRAPH 2018 Courses (SIGGRAPH '18)*. Association for Computing Machinery, New York, NY, USA, 1–94. <https://doi.org/10.1145/3214834.3214869>
- [27] Steven Schmidt, Patrick Ehrenbrink, Benjamin Weiss, Jan-Niklas Voigt-Antons, Tanja Kojic, Andrew Johnston, and Sebastian Möller. 2018. Impact of Virtual Environments on Motivation and Engagement During Exergames. In *2018 Tenth International Conference on Quality of Multimedia Experience (QoMEX)*. 1–6. <https://doi.org/10.1109/QoMEX.2018.8463389> ISSN: 2472-7814.
- [28] Dominik Schön, Thomas Kosch, Julius von Willich, Max Mühlhäuser, and Sebastian Günther. 2023. VRow-VRow-VRow-Your-Boat: A Toolkit for Integrating Commodity Ergometers in Virtual Reality Experiences. In *Proceedings of the 22nd International Conference on Mobile and Ubiquitous Multimedia (MUM '23)*. Association for Computing Machinery, New York, NY, USA, 489–491. <https://doi.org/10.1145/3626705.3631785>
- [29] Nur Aqilah Shuib, Mohd Shahrizal Sunar, Nurshamine Nazira Mohd Nor, Azizul Azman, Mohd Najeb Jamaludin, and Hadafi Fitri Mohd Latip. 2020. Rowing Simulation using Rower Machine in Virtual Reality. In *2020 6th International Conference on Interactive Digital Media (ICIDM)*. 1–6. <https://doi.org/10.1109/ICIDM51048.2020.9339603>
- [30] Ivan E. Sutherland. 1965. The ultimate display. *Multimedia: From Wagner to virtual reality* 2 (1965), 506–508.
- [31] Ancrét Szpak, Stefan Carlo Michalski, and Tobias Loetscher. 2020. Exergaming With Beat Saber: An Investigation of Virtual Reality Aftereffects. *Journal of Medical Internet Research* 22, 10 (Oct. 2020), e19840. <https://doi.org/10.2196/19840> Company: Journal of Medical Internet Research Distributor: Journal of Medical Internet Research Institution: Journal of Medical Internet Research Label: Journal of Medical Internet Research Publisher: JMIR Publications Inc., Toronto, Canada.
- [32] Robby van Delden, Sascha Bergsma, Koen Vogel, Dees Postma, Randy Klaassen, and Dennis Reidsma. 2020. VR4VRT: Virtual Reality for Virtual Rowing Training. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '20)*. Association for Computing Machinery, New York, NY, USA, 388–392. <https://doi.org/10.1145/3383668.3419865>
- [33] Eduardo Velloso, Dominik Schmidt, Jason Alexander, Hans Gellersen, and Andreas Bulling. 2015. The Feet in Human-Computer Interaction: A Survey

- of Foot-Based Interaction. *Comput. Surveys* 48, 2 (Sept. 2015), 21:1–21:35. <https://doi.org/10.1145/2816455>
- [34] Julius von Willich, Martin Schmitz, Florian Müller, Daniel Schmitt, and Max Mühlhäuser. 2020. Podoportation: Foot-Based Locomotion in Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376626>
- [35] Soojeong Yoo, Phillip Gough, and Judy Kay. 2020. Embedding a VR Game Studio in a Sedentary Workplace: Use, Experience and Exercise Benefits. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376371>
- [36] Nan Zeng, Zachary Pope, and Zan Gao. 2017. Acute Effect of Virtual Reality Exercise Bike Games on College Students' Physiological and Psychological Outcomes. *Cyberpsychology, Behavior, and Social Networking* 20, 7 (July 2017), 453–457. <https://doi.org/10.1089/cyber.2017.0042> Publisher: Mary Ann Liebert, Inc., publishers.
- [37] Yayong Zhang, Bernhard E. Riecke, Thecla Schiphorst, and Carman Neustaedter. 2019. Perch to Fly: Embodied Virtual Reality Flying Locomotion with a Flexible Perching Stance. In *Proceedings of the 2019 on Designing Interactive Systems Conference (DIS '19)*. Association for Computing Machinery, New York, NY, USA, 253–264. <https://doi.org/10.1145/3322276.3322357>